

## 4.2 Mean and variance of quadratic forms

### 4.2.1 The mean needs no normality

Theorem 2.3.1 computed  $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$  for any random vector with mean  $\boldsymbol{\mu}$  and covariance  $\boldsymbol{\Sigma}$ :

$$E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}. \quad (4.2.1)$$

The formula has two parts with distinct roles. The trace term is what the noise contributes. The term  $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$  is what the signal contributes. For nonnegative definite  $\mathbf{A}$  it vanishes exactly when  $\mathbf{A} \boldsymbol{\mu} = \mathbf{0}$ , which describes the mean vectors a well-designed estimator is meant to ignore.

**Example. (Two estimators of  $\sigma^2$  under a drifting mean)**

Let  $Y_1, \dots, Y_n$  be uncorrelated with common variance  $\sigma^2$ , and consider

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2, \quad Q = \frac{1}{2(n-1)} \sum_{i=1}^{n-1} (Y_{i+1} - Y_i)^2,$$

the sample variance and the **mean square successive difference**. In matrix form  $S^2 = \mathbf{Y}^\top \mathbf{C} \mathbf{Y} / (n-1)$  with  $\mathbf{C} = \mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top$ , and  $Q = \mathbf{Y}^\top \mathbf{D}^\top \mathbf{D} \mathbf{Y} / \{2(n-1)\}$ , where  $\mathbf{D}$  is the  $(n-1) \times n$  first-difference matrix with rows  $(\dots, -1, 1, \dots)$ . Both matrices annihilate  $\mathbf{1}$ , so both estimators ignore a constant mean. Since  $\text{tr}(\mathbf{C}) = n-1$  and  $\text{tr}(\mathbf{D}^\top \mathbf{D}) = \text{tr}(\mathbf{D} \mathbf{D}^\top) = 2(n-1)$ , Eq. 4.2.1 with  $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$  shows that both are unbiased when the mean is constant.

Now let the mean drift linearly,  $\mu_i = \mu_0 + \theta i$ . Then  $\mathbf{C} \boldsymbol{\mu}$  is the centred trend and  $\mathbf{D} \boldsymbol{\mu} = \theta \mathbf{1}_{n-1}$ , so

$$E(S^2) = \sigma^2 + \theta^2 \frac{n(n+1)}{12}, \quad E(Q) = \sigma^2 + \frac{\theta^2}{2}.$$

The bias of  $S^2$  grows quadratically with the length of the series, while that of  $Q$  does not grow at all. With  $n = 20$ ,  $\sigma = 1$  and  $\theta = 0.1$ , the means are 1.350 and 1.005. Differencing removes smooth trends, and this is why  $Q$  was proposed for series whose level may wander (von Neumann 1941). The price of robustness to drift is paid in variance, which we compute below.

### 4.2.2 A canonical representation

Higher moments and distributions need normality, and one representation answers all such questions at once.

#### Theorem 4.2.1 — Canonical form of a normal quadratic form

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  and let  $\mathbf{A}$  be symmetric. Write  $\boldsymbol{\Sigma} = \mathbf{L} \mathbf{L}^\top$  with  $\mathbf{L}$  of size  $n \times k$ , and take a spectral decomposition  $\mathbf{L}^\top \mathbf{A} \mathbf{L} = \mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^\top$  with  $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_k)$ . Put

$\mathbf{b} = \mathbf{Q}^\top \mathbf{L}^\top \mathbf{A} \boldsymbol{\mu}$ . Then

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \stackrel{d}{=} \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} + \sum_{j=1}^k (\lambda_j W_j^2 + 2b_j W_j), \quad \mathbf{W} \sim N_k(\mathbf{0}, \mathbf{I}_k). \quad (4.2.2)$$

For  $\lambda_j \neq 0$  the  $j$ th term equals  $\lambda_j(W_j + b_j/\lambda_j)^2 - b_j^2/\lambda_j$ . The nonzero  $\lambda_j$  are the nonzero eigenvalues of  $\mathbf{A}\boldsymbol{\Sigma}$ .

*Proof.* Such an  $\mathbf{L}$  exists. By Theorem 1.7.2,  $\boldsymbol{\Sigma} = \mathbf{U}\mathbf{D}\mathbf{U}^\top$  with  $\mathbf{U}$  orthogonal and  $\mathbf{D}$  diagonal with entries  $d_i \geq 0$ , and  $\mathbf{L} = \mathbf{U}\mathbf{D}^{1/2}$  works with  $k = n$ . Let  $\mathbf{Z} \sim N_k(\mathbf{0}, \mathbf{I})$ . The vector  $\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$  is normal with mean  $\boldsymbol{\mu}$  and covariance  $\mathbf{L}\mathbf{L}^\top = \boldsymbol{\Sigma}$  (Theorem 3.3.1). A normal distribution is determined by its mean and covariance (Theorem 3.2.1), so  $\mathbf{Y}$  and  $\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$  have the same law, and so do  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  and

$$(\boldsymbol{\mu} + \mathbf{L}\mathbf{Z})^\top \mathbf{A} (\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}) = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} + 2\boldsymbol{\mu}^\top \mathbf{A} \mathbf{L} \mathbf{Z} + \mathbf{Z}^\top \mathbf{L}^\top \mathbf{A} \mathbf{L} \mathbf{Z}.$$

Put  $\mathbf{W} = \mathbf{Q}^\top \mathbf{Z}$ , again  $N_k(\mathbf{0}, \mathbf{I})$ . Then  $\mathbf{Z}^\top \mathbf{L}^\top \mathbf{A} \mathbf{L} \mathbf{Z} = \mathbf{W}^\top \boldsymbol{\Lambda} \mathbf{W} = \sum \lambda_j W_j^2$  and  $\boldsymbol{\mu}^\top \mathbf{A} \mathbf{L} \mathbf{Z} = \mathbf{b}^\top \mathbf{W}$ , which is Eq. 4.2.2. The completed square is algebra. Finally,  $\mathbf{L}^\top \mathbf{A} \mathbf{L} = \mathbf{H}\mathbf{F}$  with  $\mathbf{F} = \mathbf{A}\mathbf{L}$  and  $\mathbf{H} = \mathbf{L}^\top$ , and  $\mathbf{F}\mathbf{H} = \mathbf{A}\boldsymbol{\Sigma}$ . By Proposition 1.7.1(d) these two products have the same nonzero eigenvalues, with the same multiplicities. ■

So a normal quadratic form is a constant plus a sum of independent pieces. Each piece is either a scaled noncentral  $\chi^2(1)$  variable, with scale  $\lambda_j$ , or a normal variable with mean zero, when  $\lambda_j = 0$  but  $b_j \neq 0$ . Because the  $\lambda_j$  may be negative,  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  need not be positive. The linear pieces are special to singular covariances. If  $\boldsymbol{\Sigma}$  is positive definite,  $\mathbf{L}$  can be taken square and nonsingular, and then  $\lambda_j = 0$  gives  $\mathbf{A}\mathbf{L}\mathbf{q}_j = \mathbf{0}$  for the  $j$ th column  $\mathbf{q}_j$  of  $\mathbf{Q}$ , so  $b_j = (\mathbf{A}\mathbf{L}\mathbf{q}_j)^\top \boldsymbol{\mu} = 0$ . A linear piece needs a direction in which  $\mathbf{A}$  responds to the mean while the noise is absent.

### Corollary 4.2.2 — Moment generating function

In the setting of Theorem 4.2.1, for  $|t| < 1/(2 \max_j |\lambda_j|)$  (any  $t$  if all  $\lambda_j = 0$ ),

$$\mathbb{E} e^{t\mathbf{Y}^\top \mathbf{A} \mathbf{Y}} = e^{t\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}} \prod_{j=1}^k (1 - 2\lambda_j t)^{-1/2} \exp\left(\frac{2b_j^2 t^2}{1 - 2\lambda_j t}\right).$$

*Proof.* The terms of Eq. 4.2.2 are independent. If  $\lambda_j = 0$ , the term  $2b_j W_j$  has mgf  $e^{2b_j^2 t^2}$ . If  $\lambda_j \neq 0$ , apply Lemma 4.1.1 to  $W_j + \nu_j$  with  $\nu_j = b_j/\lambda_j$  at the argument  $\lambda_j t$ , and multiply by  $e^{-tb_j^2/\lambda_j}$ . The exponent is  $b_j^2 t / \{\lambda_j(1 - 2\lambda_j t)\} - b_j^2 t / \lambda_j = 2b_j^2 t^2 / (1 - 2\lambda_j t)$ . ■

## 4.2.3 Variance and covariances

**Theorem 4.2.3 — Moments of normal quadratic forms**

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ , let  $\mathbf{A}$  and  $\mathbf{B}$  be symmetric  $n \times n$  matrices, and let  $\mathbf{K}$  be an  $m \times n$  matrix. Then

- (a)  $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ ;
- (b)  $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 2 \text{tr}\{(\mathbf{A} \boldsymbol{\Sigma})^2\} + 4 \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}$ ;
- (c)  $\text{Cov}(\mathbf{K} \mathbf{Y}, \mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 2 \mathbf{K} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}$ ;
- (d)  $\text{Cov}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}, \mathbf{Y}^\top \mathbf{B} \mathbf{Y}) = 2 \text{tr}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{B} \boldsymbol{\Sigma}) + 4 \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{B} \boldsymbol{\mu}$ .

*Proof.*

- (a) is Eq. 4.2.1. For (b), use Eq. 4.2.2. Independent terms have additive variances, and for  $W \sim N(0, 1)$ ,  $\text{Var}(\lambda W^2 + 2bW) = 2\lambda^2 + 4b^2$ , because  $\text{Cov}(W^2, W) = E W^3 = 0$ . Hence

$$\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 2 \sum_j \lambda_j^2 + 4 \|\mathbf{b}\|^2 = 2 \text{tr}\{(\mathbf{L}^\top \mathbf{A} \mathbf{L})^2\} + 4 \boldsymbol{\mu}^\top \mathbf{A} \mathbf{L} \mathbf{Q} \mathbf{Q}^\top \mathbf{L}^\top \mathbf{A} \boldsymbol{\mu}.$$

Now  $\text{tr}\{(\mathbf{L}^\top \mathbf{A} \mathbf{L})^2\} = \text{tr}(\mathbf{L}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \mathbf{L}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma})$  by Theorem 1.6.1, and  $\mathbf{Q} \mathbf{Q}^\top = \mathbf{I}$ , so the second term is  $4 \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}$ . For (c), write  $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{L} \mathbf{Z}$  as in the proof of Theorem 4.2.1. Then  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} - E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \mathbf{Z}^\top \mathbf{L}^\top \mathbf{A} \mathbf{L} \mathbf{Z} - \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + 2 \boldsymbol{\mu}^\top \mathbf{A} \mathbf{L} \mathbf{Z}$  and  $\mathbf{K} \mathbf{Y} - E(\mathbf{K} \mathbf{Y}) = \mathbf{K} \mathbf{L} \mathbf{Z}$ . Every entry of  $E[\mathbf{Z}(\mathbf{Z}^\top \mathbf{G} \mathbf{Z})]$  is a sum of third moments of a  $N_k(\mathbf{0}, \mathbf{I})$  vector, which vanish. So only the linear term contributes, and  $E[\mathbf{K} \mathbf{L} \mathbf{Z} \cdot 2 \mathbf{Z}^\top \mathbf{L}^\top \mathbf{A} \boldsymbol{\mu}] = 2 \mathbf{K} \mathbf{L} \mathbf{L}^\top \mathbf{A} \boldsymbol{\mu}$ . For (d), apply (b) to  $\mathbf{A}$ ,  $\mathbf{B}$  and  $\mathbf{A} + \mathbf{B}$ , and use  $\text{Var}(X + Y) = \text{Var} X + \text{Var} Y + 2 \text{Cov}(X, Y)$ . ■

For  $\boldsymbol{\mu} = \mathbf{0}$ , part (b) can also be read off the fourth moments of a normal vector (Exercise (B1)). The variance has the same two-part structure as the mean: a pure-noise term, plus a term that is present only when  $\mathbf{A}$  responds to the mean. Part (c) shows that a quadratic form and a linear form are uncorrelated when  $\mathbf{K} \boldsymbol{\Sigma} \mathbf{A} = \mathbf{0}$ . Under normality this will turn out to give independence (Theorem 4.4.1).

**Example.****Example (Two estimators of  $\sigma^2$  under a drifting mean)**

With  $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$  and a constant mean, Theorem 4.2.3(b) gives  $\text{Var}(S^2) = 2\sigma^4 \text{tr}(\mathbf{C}^2)/(n-1)^2 = 2\sigma^4/(n-1)$ . For  $Q$ , the matrix  $\mathbf{D} \mathbf{D}^\top$  is tridiagonal with 2 on the diagonal and -1 beside it, so  $\text{tr}\{(\mathbf{D}^\top \mathbf{D})^2\} = \text{tr}\{(\mathbf{D} \mathbf{D}^\top)^2\} = 4(n-1) + 2(n-2) = 6n-8$  and

$$\text{Var}(Q) = \frac{2\sigma^4(6n-8)}{4(n-1)^2} = \frac{\sigma^4(3n-4)}{(n-1)^2}.$$

The ratio  $\text{Var}(Q)/\text{Var}(S^2) = (3n-4)/\{2(n-1)\}$  tends to 3/2. With  $n = 20$  the two variances are 0.1551 and 0.1053, a ratio of 1.474. So with no drift,  $Q$  needs about half as many observations again to match  $S^2$ . Under the drift  $\theta = 0.1$ , the signal term  $4 \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}$  raises the variances to 0.1552 and 0.1789. Combining variance and bias, the

root mean squared errors are 0.394 for  $Q$  and 0.549 for  $S^2$ . The listing evaluates the formulas, and the script confirms all eight moments with 400,000 simulated series.

```
import numpy as np
rng = np.random.default_rng(4402)
n, sigma, theta = 20, 1.0, 0.1
t = np.arange(1, n + 1)
mu = 5.0 + theta * t                                # mean with a linear trend
Sigma = sigma**2 * np.eye(n)

def qf_moments(A, mu, Sigma):
    """Mean and variance of  $Y^T A Y$  for  $Y \sim N(\mu, \text{Sigma})$ ,  $A$  symmetric."""
    AS = A @ Sigma
    mean = np.trace(AS) + mu @ A @ mu
    var = 2 * np.trace(AS @ AS) + 4 * mu @ AS @ A @ mu
    return mean, var

C = np.eye(n) - np.ones((n, n)) / n                # centring matrix
D = np.diff(np.eye(n), axis=0)                      # (n-1) x n first differences
A_s2 = C / (n - 1)
A_q = D.T @ D / (2 * (n - 1))
for name, A in [("S2", A_s2), ("Q", A_q)]:
    print(name, qf_moments(A, mu, Sigma), qf_moments(A, 0 * mu, Sigma))
```

[Run this code online](#)

Normality enters (b)–(d) through the third and fourth moments of  $\mathbf{Y}$ . For independent coordinates with excess kurtosis the variance acquires an extra term that involves only the diagonal of  $\mathbf{A}$  (Theorem 2.3.3). Variance formulas for sums of squares derived under normality can be badly wrong for heavy-tailed errors, a point taken up in Chapter 21.

## 4.2.4 Exercises

### 4.2.4.1 A. Check your understanding

#### Exercise A1.

Let  $\mathbf{Y} \sim N_2(\mathbf{0}, \mathbf{I})$  and  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ . Compute  $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$  directly, and compare with  $2 \text{tr}(\mathbf{A}^2)$  and with  $2 \text{tr}(\mathbf{A}_s^2)$ , where  $\mathbf{A}_s = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$ .

### 4.2.4.2 B. Practice

#### Exercise B1.

Let  $(U_i, W_i), i = 1, \dots, n$ , be independent copies of a bivariate normal pair with variances  $\sigma_U^2, \sigma_W^2$  and covariance  $\sigma_{UW}$ , and let  $s_{UW} = (n - 1)^{-1} \sum_i (U_i - \bar{U})(W_i - \bar{W})$ . Write  $s_{UW} = \mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  for the stacked vector  $\mathbf{Y} = (\mathbf{U}^\top, \mathbf{W}^\top)^\top$  and a symmetric  $2n \times 2n$  matrix  $\mathbf{A}$ , and use Theorem 4.2.3 to show that  $\text{Var}(s_{UW}) = (\sigma_U^2 \sigma_W^2 + \sigma_{UW}^2) / (n - 1)$ . Check the answer against  $\text{Var}(S^2)$  when  $\mathbf{U} = \mathbf{W}$ .

**Exercise B2.**

Let  $Y_1, \dots, Y_n$  be independent with mean 0, variance  $\sigma^2$  and  $EY_i^4 = \mu_4$ . By Exercise (B4), for symmetric  $\mathbf{A}$  and  $\mathbf{B}$ ,

$$\text{Cov}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}, \mathbf{Y}^\top \mathbf{B} \mathbf{Y}) = (\mu_4 - 3\sigma^4) \sum_i a_{ii} b_{ii} + 2\sigma^4 \text{tr}(\mathbf{A} \mathbf{B}).$$

Show that in a one-way layout with  $g \geq 2$  groups, not all of size one, the between-group and within-group sums of squares are correlated unless  $\mu_4 = 3\sigma^4$ . Which sign does the correlation have for heavy-tailed errors?

*Solution.* In the one-way layout the between-group matrix is  $\mathbf{A} = \mathbf{M} - \mathbf{P}_1$  and the within-group matrix is  $\mathbf{B} = \mathbf{I} - \mathbf{M}$ . They satisfy  $\mathbf{A} \mathbf{B} = \mathbf{0}$ , so  $\text{tr}(\mathbf{A} \mathbf{B}) = 0$  and the covariance reduces to  $(\mu_4 - 3\sigma^4) \sum_i a_{ii} b_{ii}$ . An observation in group  $\ell$  has  $a_{ii} = 1/n_\ell - 1/n$  and  $b_{ii} = 1 - 1/n_\ell$ , so

$$\sum_i a_{ii} b_{ii} = \sum_\ell n_\ell \left( \frac{1}{n_\ell} - \frac{1}{n} \right) \left( 1 - \frac{1}{n_\ell} \right) = \sum_\ell \left( 1 - \frac{n_\ell}{n} \right) \left( 1 - \frac{1}{n_\ell} \right).$$

With  $g \geq 2$  every first factor is positive, and the second is positive for any group with two or more observations. So the sum is positive, and the covariance vanishes iff  $\mu_4 = 3\sigma^4$ . For heavy-tailed errors,  $\mu_4 > 3\sigma^4$ , the two sums of squares are positively correlated: a large error inflates both.

**Exercise B3.**

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$ , and let  $\mathbf{M}$  be symmetric idempotent of rank  $r$  with  $\mathbf{M} \boldsymbol{\mu} \neq \boldsymbol{\mu}$ , as when a regression model omits a needed regressor. Show that  $\|(\mathbf{I} - \mathbf{M}) \mathbf{Y}\|^2 / \sigma^2 \sim \chi^2(n - r, \|(\mathbf{I} - \mathbf{M}) \boldsymbol{\mu}\|^2 / \sigma^2)$ , and find the bias of  $\|(\mathbf{I} - \mathbf{M}) \mathbf{Y}\|^2 / (n - r)$  as an estimator of  $\sigma^2$ . In the setting of Theorem 4.4.6 without the assumption  $\mathbf{P} \boldsymbol{\mu} = \boldsymbol{\mu}$ , which parts of the conclusion survive?

**Exercise B4.**

Let  $\mathbf{Y} \sim N_n(\mathbf{0}, \boldsymbol{\Sigma})$ . Use Corollary 4.2.2 and Proposition 1.7.1(d) to show that  $E e^{t \mathbf{Y}^\top \mathbf{A} \mathbf{Y}} = \det(\mathbf{I}_n - 2t \mathbf{A} \boldsymbol{\Sigma})^{-1/2}$  for  $|t|$  small.

**4.2.4.3 C. Going deeper****Exercise C1.**

Let  $\mathbf{Y} \sim N_n(\mu \mathbf{1}, \sigma^2 \mathbf{I})$  with  $\mu$  and  $\sigma^2$  unknown. Among symmetric  $\mathbf{A}$  with  $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2$  for all  $\mu$  and  $\sigma^2$ , show that  $\mathbf{A} = \mathbf{C}/(n - 1)$ , which gives  $S^2$ , uniquely minimizes  $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$  at every parameter value. (Compare Exercise (B5), where the mean is known to be zero.)

*Solution.* Since  $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2 \text{tr} \mathbf{A} + \mu^2 \mathbf{1}^\top \mathbf{A} \mathbf{1}$ , unbiasedness means  $\text{tr} \mathbf{A} = 1$  and  $\mathbf{1}^\top \mathbf{A} \mathbf{1} = 0$ . By Theorem 4.2.3,  $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 2\sigma^4 \text{tr}(\mathbf{A}^2) + 4\mu^2 \sigma^2 \|\mathbf{A} \mathbf{1}\|^2$ . The map  $\mathbf{A} \mapsto \mathbf{C} \mathbf{A} \mathbf{C}$  is an orthogonal projection for the inner product  $\text{tr}(\mathbf{A} \mathbf{B})$  on symmetric

matrices, so  $\text{tr}(\mathbf{A}^2) \geq \text{tr}\{(\mathbf{CAC})^2\}$ , with equality iff  $\mathbf{A} = \mathbf{CAC}$ . Also  $\text{tr}(\mathbf{CAC}) = \text{tr } \mathbf{A} - \mathbf{1}^\top \mathbf{A} \mathbf{1} / n = 1$ . The symmetric matrix  $\mathbf{CAC}$  has rank at most  $n - 1$ . If  $\lambda_1, \dots, \lambda_{n-1}$  are its eigenvalues on  $\mathbf{1}^\perp$ , the Cauchy-Schwarz inequality gives  $1 = (\sum \lambda_i)^2 \leq (n - 1) \sum \lambda_i^2$ . Hence  $\text{tr}(\mathbf{A}^2) \geq 1/(n - 1)$ , with equality iff  $\mathbf{A} = \mathbf{CAC}$  and all  $\lambda_i = 1/(n - 1)$ , that is, iff  $\mathbf{A} = \mathbf{C}/(n - 1)$ . That matrix also has  $\mathbf{A} \mathbf{1} = \mathbf{0}$ , so it minimizes both variance terms at once.